Qiskit Fall Fest 2026

Challenge 4: Quantum Correlation Switch — Complete Solution

This solution follows the uploaded Challenge 4 PDF (page 1): two qubits, measurement of both, at most four gates before measurement, and 100 shots for each experiment.

1. Part A — Correlated outcomes

Prediction: H on qubit 0 followed by CNOT with control 0 and target 1 prepares (|00■ + |11■)/√2. The dominant outcomes are 00 and 11, ideally 50% each.

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_histogram
import matplotlib.pyplot as plt

simulator = AerSimulator()
part_a = QuantumCircuit(2, 2)
part_a.h(0)
part_a.cx(0, 1)
part_a.measure([0, 1], [0, 1])
result_a = simulator.run(transpile(part_a, simulator), shots=100).result()
counts_a = result_a.get_counts()
print(counts_a)
plot_histogram(counts_a); plt.show()
```

2. Part B — Opposite outcomes

Modification: Add one X gate to qubit 0 after preparing the Bell state. The resulting state is (|01■ + |10■)/√2, so the dominant outcomes are 01 and 10.

```
part_b = QuantumCircuit(2, 2)
part_b.h(0)
part_b.cx(0, 1)
part_b.x(0)
part_b.measure([0, 1], [0, 1])
result_b = simulator.run(transpile(part_b, simulator), shots=100).result()
counts_b = result_b.get_counts()
print(counts_b)
plot_histogram(counts_b); plt.show()
```

3. Comparison

Part	Gates	Expected outcomes	Relationship
A	H, CNOT	00 and 11	Correlated
B	H, CNOT, X	01 and 10	Anti-correlated

4. Explanation and conclusion

In Part A, H creates a superposition and CNOT entangles the qubits, producing (|00■ + |11■)/√2. Therefore, the measured bits match. In Part B, the extra X gate flips one qubit and changes the state to (|01■ + |10■)/√2, so the measured bits are opposite. An ideal simulator produces only the two expected outcomes for each circuit. Real hardware may show small counts for other outcomes due to noise and readout errors. Counts for the two allowed outcomes should be close to 50 each over 100 shots, but need not be exactly equal.

Note: Qiskit was not installed in the preparation environment. The provided code generates actual 100-shot counts and histograms when run with Qiskit and Qiskit Aer installed.